2026 Spring VLSI DSP Homework Assignment #2

Due date: 2026/4/14

Q1. Following the 2-D DWT algorithm given in homework assignment 1, please conduct fixed point simulations to determine the DWT word length for the following items. Assume a floating point version IDWT is used in image reconstruction, the PSNR (peak signal to noise ratio) of the reconstructed image should be no less than 40dB. Please use as small word length as possible to achieve this goal. You need to indicate both the "word length" and the "weight" of LSB of each variable.

- The word length of the filter coefficients (Table 1), all coefficients should have the same word length. You can determine whether they will share the same weight or not. If they share the same weight, the effective word length of smaller coefficients will be reduced. However, there is no need of weight alignment when performing addition in filtering.

Analysis Filter Coefficients		
i	Lowpass Filter h_i	Highpass Filter g_i
0	0.852698679009	-0.788485616406
±1	0.377402855613	0.418092273222
±2	-0.110624404418	0.040689417609
±3	-0.023849465020	-0.064538882629
±4	0.037828455507	

- Determine the word lengths (and the weights) of the filter outputs at each level, i.e., level 1, 2 and 3 of DWT.
- **Note 1:** For 2-D DWT, you need to perform 1-D DWTs horizontally first, followed by performing 1-D DWTs vertically using the results of horizontal DWTs. Therefore, the word length designs for horizontal and vertical DWTs can be different.
- Please summarize your final <u>bit-true</u> algorithm design by listing the word length and weight of each variable in 2-D DWT, and calculate the PSNR of reconstructed image.
- **Note 2:** reconstructed image is obtained by performing <u>fixed point</u> forward 2-D DWT followed by performing <u>floating point</u> inverse 2-D DWT

Q2. Develop a hardware design of 2-D DWT bit true algorithm you derive in Q1. Employ the architecture shown in Fig. 1.

- **Note 3:** You don't need to calculate every input sample and then discard every other output, i.e. down sample the results. Instead, you should calculate the useful outputs only.

- You need a temporary memory storage (transpose memory) to keep the result of horizontal DWTs before the start of vertical DWTs. Please design your memory configuration, i.e. how many memory modules and the depth and width of each memory module.
- **Note 4:** If you perform vertical DWTs only **after** all horizontal DWTs are completed, it means you need to buffer almost the entire 512X512 image in your on-chip memory modules. The area overhead would be too high. Refer to Fig. 2, a better way of designing your transpose memory would be would be buffer just a few lines of horizontal DWT results and start your vertical DWT right after you have sufficient input samples, i.e., 9 samples for low pass filtering.
- After 1st level of decomposition, you can write HL, LH and HH sub-bands data to external memory to reduce the on-chip memory size. For simplicity, you don't have to implement this part in your design.
- Please code your design in Verilog and conduct simulation to verify your design. Use the fixed-point simulation results obtained in Q1 as the golden pattern of testbench. Your Verilog simulation results should be identical to fixed point simulation results.
- Synthesize your design and report the following items. You may choose either TSMC ADFP virtual process or AMD (Xilinx) FPGA XC7Z020 as your target technology.
  1) For cell based design, report equivalent gate count or area, please also indicate the percentages of combinational circuit (filter) and memory modules
  2) For FPGA design, give resource utilization table showing the numbers of FFs, LUTs, DSP slices, and block RAMs used in your design.
  3) Number of clock cycles to complete the entire 2-D DWT computations
  4) Maximum working frequency

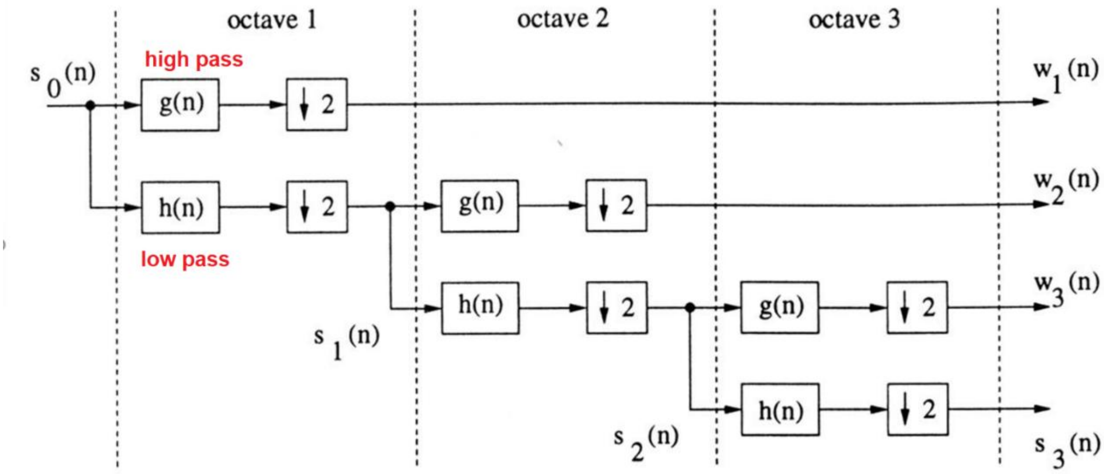

Fig. 1 architecture of 1-D 3-level DWT transform

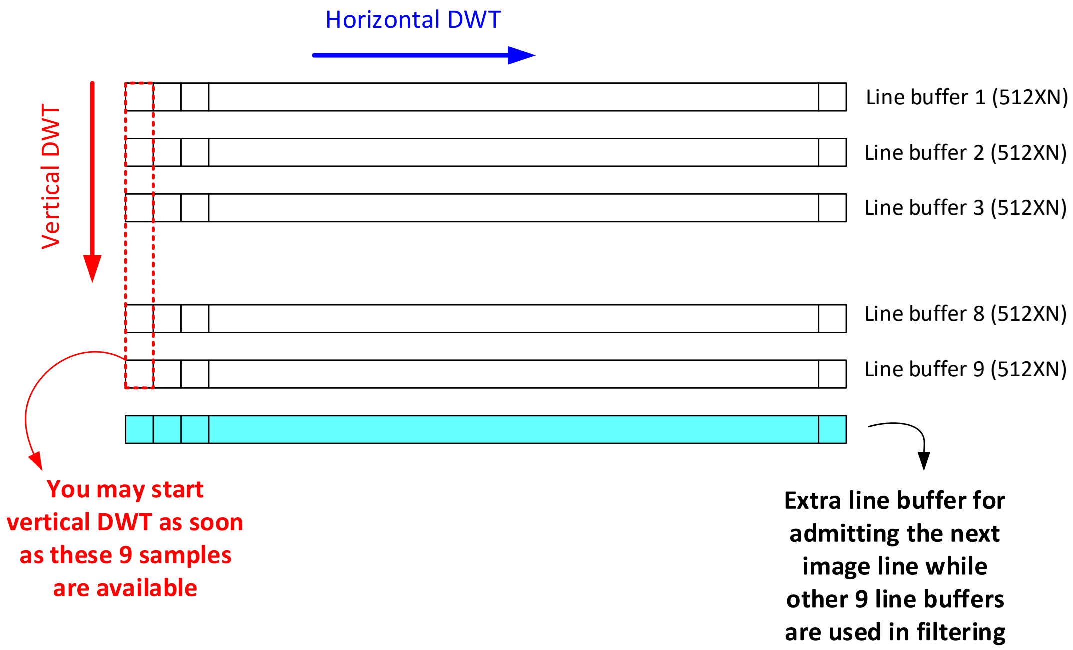

Fig. 2 Efficient computing scheme and transpose memory design